(a) Evaluation Harness Infrastructure

Sec. 8.1 | Execution Substrate, Telemetry, and Replay

1. Campaign Input Workloads

Attack Engines
Adaptive search and mutation
High-throughput exploit generation
GPTFuzzer, APRT, TAP

Evaluation Suites
Standardized benchmark suites
Controlled comparative baselines
HarmBench, AgentDojo

2. Core Harness Substrate (garak, PyRIT, Inspect)

Target Adapters
Model API wrappers and runners
promptfoo, garak

Attack Orchestrator
Multi-turn state and budget
PyRIT, ARTKIT, EasyJailbreak

Execution Runtime and Sandbox
Isolated tool sandbox and trace logging
Inspect AI sandbox, AgentDojo

Automated Judges and Scorers
Model-graded rubrics, refusal scoring, assertions
WildGuard, StrongREJECT, LlamaGuard

3. Evidence Preservation and Replay

Immutable Audit Traces
Run configs, seeds, and logs
Judge reasoning and verdicts
Structured telemetry state logs
Inspect AI logs, PyRIT DuckDB

Deterministic Replay
Fixed-seed target regression
Automated safety drift checks
Cross-version benchmark parity
promptfoo eval, AgenticEval

Immutable traces preserve experimental evidence for reproducible auditing

(b) Human-in-the-Loop Synergies

Sec. 8.2 | Human Specification and Algorithmic Scaling

1. Human Qualitative Specification

Normative Scope
Harm boundaries and safety redlines
STAR, Alvarado G.

Cultural Context
Localized norms and contextual cues
Going PLACES

Disagreement Models
Annotator variance
Boundary arbitration
DICES, D3CODE

Qualitative expertise grounds risk taxonomies and resolves boundary ambiguity

2. Algorithmic Scaling and Batch Screening

Workload Multiplier
Recipe-guided prompt mutation
Sociolinguistic diversity scaling
Expands coverage to 10k+ variants
AART, Bridging Scale Gap

Automated Batch Triage
High-throughput batch screening
Cluster analysis isolates flaws
Filters passes; isolates edge cases
WildGuard, StrongREJECT

Batch screening routes edge cases to experts

3. Expert Adjudication and Rubric Recalibration

Criteria and Rubric Calibration
Update rubrics, classification thresholds
Sync versioned ground-truth criteria
STAR, Constitutional AI feedback

Verdicts

Human Expert Adjudication
Arbitrate annotator disagreement
Isolate emergent vulnerability types
DICES, D3CODE, Human Panels

Risk Register and Evaluation Standards Refinement
Adjudicated boundary cases update risk registers and escalation thresholds
STAR, Alvarado Garcia et al., Going PLACES

Expert adjudication isolates novel harms, driving continuous rubric evolution

(c) Artifact Governance and Multi-Party CVD

Sec. 8.3 | Balancing Transparency and Downstream Exploitability

1. Tiered Artifact Access Shelves

Tier 1 (Public): Open benchmark suites and leaderboards (HarmBench)

Tier 2 (Gated): Controlled access for vetted researchers (Fang et al.)

Tier 3 (Withheld): Exploits withheld; system cards published (OpenAI)

Artifact access is tiered by dual-use risk and downstream exploitability

2. Finding Accountability Reporting Standard

Threat Model and Setup
Model version, wrapper, query budget

Evaluation Protocol
Seeds, step limits, judge rubrics

Evidence and Artifacts
Redacted traces and proof package

Disclosure Lifecycle
Confidential notice, patch, advisory

Standardized dimensions enable verifiable cross-party security auditing

3. Coordinated Vulnerability Disclosure (CVD) *(FLARE-AI, CERT/CC)*

Red-Team Security Auditor

1. Defect and Proof Package
Submit redacted PoC and traces

Target Model Vendor

2. Flaw Triage
Verify severity

3. Patch Dev
Filters and fine-tuning

Coordinator and Community

4. Joint Security Advisory
Advisory released upon patch validation

Coordinated disclosure pipeline validates mitigations prior to advisory